# Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

## Answer 1 of 7 · Question 4 part b.iii · 4 marks

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 4 | (a) | (i) | • decreases;<br>• from 82% to 72%/by 10% | | 2 |
| | | (ii) | 79:21;<br>3.76:1; | accept 3.7612:1<br>accept 3.762:1<br>accept 3.8/4:1 | 2 |
| | | (iii) | any two from<br>• passive smoking/ex- smokers already have damaged lungs;<br>• (air) pollution;<br>• genetics/heredity; | | 2 |
| | | (iv) | tar; | | 1 |
| | (b) | (i) | bronchiole; | | 1 |
| | | (ii) | alveolus; | | 1 |
| | | (iii) | any four from<br>• damaged alveoli/alveoli walls broken;<br>• less surface area (for gas exchange);<br>• difficulty breathing/shortness of breath;<br>• less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange;<br>• reduced (aerobic) respiration;<br>• fatigue/less energy/not very active; | | 4 |
| | | | | **Total for Question 4 = 13 marks** | |

# Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

## Answer 2 of 7 · Question 6 part c.ii · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 | (a) (i) | Increase/goes up; | allow equivalent | 1 |
| | (ii) | Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;; | Allow one mark for reference to lifestyles only | 2 |
| | (b) | Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed;<br>Increased risk of (lung/respiratory) infection/coughing; | | 1<br>1 |
| | (c) (i) | <table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table> | Allow for SA:V ratio 6:3 ECF | 3 |
| | (ii) | Smaller surface area to volume ratio (in lungs with emphysema);<br>Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide;<br>Less oxygenated blood/less oxygen transferred (to cells);<br>Less (aerobic) respiration; | | Max 3 |

# Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

## Answer 2 of 7 · Question 6 part c.ii · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 3 of 7 · Question 3 part a.ii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | C; (red blood cells) | | 1 |
| (ii) | any three from<br>• more carbon dioxide (1)<br>• less oxygen (1)<br>• moister (1)<br>• warmer (1) | | 3 |
| (iii) | • large surface area (1)<br>• thin (walled)/wall one cell thick (1)<br>• good blood supply (1)<br>• moist lining (1) | | 3 |
| (b) | • thin wall (1)<br>• narrow lumen (1) | Allow references to only one blood cell passing through | 2 |
| (c) | • breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1)<br>• as lungs would not squeeze out air/forced exhalation(1) | | 2 |

Total 11 marks

## Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

### Answer 3 of 7 · Question 3 part a.ii · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 4 of 7 · Question 3 part a.iii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | C; (red blood cells) | | 1 |
| (ii) | any three from<br>• more carbon dioxide (1)<br>• less oxygen (1)<br>• moister (1)<br>• warmer (1) | | 3 |
| (iii) | • large surface area (1)<br>• thin (walled)/wall one cell thick (1)<br>• good blood supply (1)<br>• moist lining (1) | | 3 |
| (b) | • thin wall (1)<br>• narrow lumen (1) | Allow references to only one blood cell passing through | 2 |
| (c) | • breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1)<br>• as lungs would not squeeze out air/forced exhalation(1) | | 2 |

Total 11 marks

# Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

## Answer 4 of 7 · Question 3 part a.iii · 3 marks

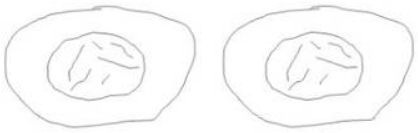
2020 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 5 of 7 · Question 2 part a.iii · 4 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 156 – 72;<br>84; | ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark | 2 |
| (ii) | **C** 9 minutes; | | 1 |
| (iii) | In the following order only:<br>• oxygen;<br>• diffusion;<br>• aerobic/cellular;<br>• decreases/reduces/goes down; | | 4 |
| (b) (i) |  <br>must have 6 chromosomes shape not important | One mark for each correct cell drawn | 2 |
| (ii) | **B** metaphase;<br>A is incorrect as chromosomes do not align at the equator during anaphase<br>C is incorrect as chromosomes do not align at the equator during prophase<br>D is incorrect as chromosomes do not align at the equator during telophase | | 1 |
| | | | Total 10 |

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## Answer 5 of 7 · Question 2 part a.iii · 4 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| uestion number | Answer | Notes | Mark s |
|---|---|---|---|

## Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

### Answer 6 of 7 · Question 2 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;; | one mark for two correct<br>two marks for four correct | 2 |
| | | (ii) | • large surface area;<br>• thin epithelium/walls; | R cell wall/cell membrane | 2 |
| | (b) | (i) | B; 0.04%<br>A, C and D are all incorrect figures | | 1 |
| | | (ii) | • $1200 \times 16$;<br>• $\frac{19200}{1000}$ = 19 $dm^3$; | ECF if any value used from table<br>Accept 19.2 | 2 |
| | | (iii) | • $\frac{20 \times 10\,500;}{100}$<br>• $\frac{2100 \times 20;}{100}$<br>• = 420 $cm^3$; | 2 mark ECF | 3 |
| | | (iv) | • more carbon dioxide in blood/lungs/body;<br>• detected by (chemo)receptors (in arteries);<br>• rate of breathing/tidal volume increases;<br>• increases to remove carbon dioxide; | | 4 |

Total 14 marks

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## Answer 6 of 7 · Question 2 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 7 of 7 · Question 4 part a.ii · 4 marks

2024 JUN · Paper 02 · find part a.ii in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 | (a) | (i) | Correct labels to:<br><br>trachea;<br>bronchus/bronchi;<br>alveoli; | (3) |
| | | (ii) | • many alveoli/large surface area;<br>• short diffusion distance/ alveoli 1 cell thick/capillary 1 cell thick/ thin wall;<br>• moist;<br>• (dense/extensive/large) network of capillaries; | (4) |
| | (b) | (i) | any four from<br>• reference to use of a nose clip;<br>• reference to soda lime ( to absorb carbon dioxide)<br>• reference to sterilising mouth piece;<br>• breathe in and out( normally) through a mouthpiece(connected to spirometer);<br>• pen goes up and down (on a rotating drum);\<br>• measure peak to trough/ highest to lowest/top to bottom/ from the trace; | (4) |
| | | (ii) | 4600 ÷ 30 (x 100) = 1380;<br>4600 – 1380 = 3220 ($cm^3$);<br>or<br>4600×0.7=3220 ($cm^3$);; | ecf: allow correct calculation from values given for mp1. Max 2 marks<br>(2) |

**Total for Question 4 = 13 marks**